Using predictive dynamics as a window into the evolution of language and music

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1. Background

Humans are unique among social animals in having both music and language (M&L). A key approach to studying M&L evolution is identifying shared traits (e.g., vocal learning, entrainment) across species and mapping them onto phylogenetic trees (Honing et al., 2015). Cross-cultural work reveals near-universal features specific to or shared by M&L (Savage et al., 2015). However, many crucial traits emerged after our split from chimpanzees, limiting explanatory power. Understanding the evolutionary sequence and interaction of these traits could refine hypotheses about protolanguage and protomusic.

2. Aims

We propose adapting an evolutionary systems biology (ESB) framework aiming to identify causal interactions in evolving systems and define their possible evolutionary trajectories. Using dynamical systems modeling, this approach provides mechanistic explanations for phenotypic development and evolutionary transitions. The key domain to assess evolvability and robustness is the component dynamics of the system (in the ESB framework: gene expression dynamics linking genotype to phenotype, Jaeger & Monk, 2021). Since prediction is the mechanism by which cognitive capacities are expressed in real-time vocal interaction, we focus on predictive dynamics as mediating between cognitive ("genotypic") traits and spectrotemporal ("phenotypic") features.

3. Framework instantiation

We reason that (1) each neurobiologically implemented cognitive capacity enables new predictive dynamics, (2) these dynamics shape the acoustic features that actualise prediction in social interactions, and (3) they can be quantified using computational and dynamical systems modeling. Because prediction can be quantified using information-theoretic measures (e.g., entropy and surprisal), this framework permits formal comparison of predictive regimes across species and proto-stages. As an instantiation of our theoretical framework, we consider the hypothesised evolutionary acquisition of beat-based temporal prediction. Treating the presence of beat inference as a parameter change in the generative model, we predict a qualitative transition in the phase portrait of predictive dynamics: from a monostable regime dominated by event-based prediction to a bistable regime

supporting oscillatory temporal expectations. This transition corresponds to the emergence of rhythmic affordances characteristic of participatory music-making. Figure 1 schematically illustrates how such a phase-portrait transition corresponds to movement in configuration space.

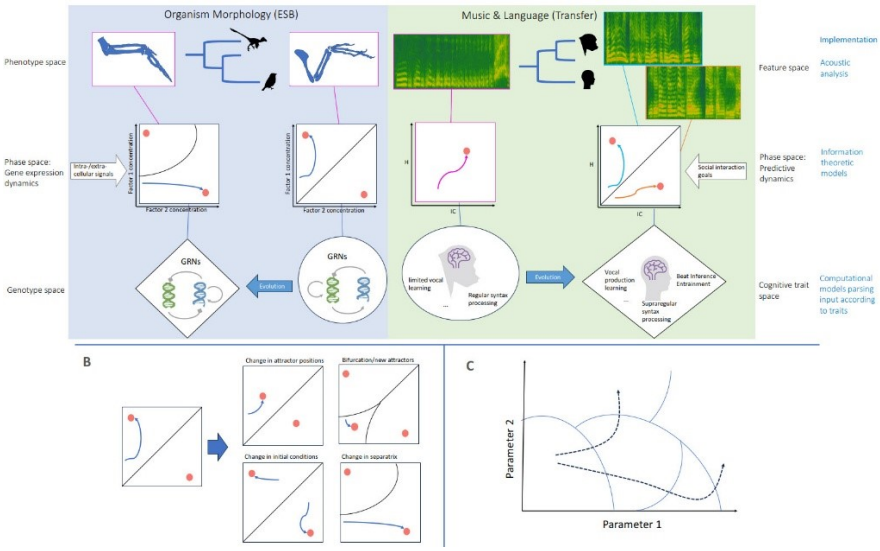

Figure 1: Overview over the ESB framework and its proposed transfer to M&L evolution.

4. Predicted dynamical regimes

Following this model template and taking the example of M&L evolution, we conceptualise dynamics of prediction as the interface mediating between a cultural phenotype such as song and speech acoustics, and underlying cognitive goals such as coordination in social interaction. Operationalising predictability in information-theoretic ways and treating key traits of musicality and language as parameters, we propose that a configuration space can be derived that could enable us to reconstruct evolutionary pathways of M&L. The framework yields testable predictions: species or cultures lacking a given cognitive trait occupy restricted regions of predictive phase space; transitions between proto-music and proto-language correspond to bifurcations in predictive dynamics rather than gradual feature drift; and similar acoustic features arise from distinct cognitive architectures, if instantiating homologous dynamical regimes.

5. Discussion and conclusion

We propose that a framework from ESB can be applied to cognition to gain empirical insight into M&L evolution. We propose equivalences between Jaeger's framework, and song and speech as cultural systems and discuss possible limitations and extensions, as concern comparative data and biosemiotics.

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